Vedavarshita Nunna

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Education

Drexel University — Philadelphia, PA
Master of Science, Artificial Intelligence & Machine Learning (CGPA: 3.9) — *Sep 2023 – June 2025*
Coursework: Intro to Cloud Computing, Data Analysis and Interpretation, Machine Learning

Lovely Professional University — Punjab, India
Bachelor of Technology, Electronics & Communication Engineering - Minor in Data Science — *July 2018 – May 2022*
Coursework: Database Management Systems, Data Science Toolbox: R Programming, Data Visualization

Experience

CODED Lab, Drexel University — July 2024 – present
AI/ML Research Assistant – Advisor: Dr. Jake Williams) | Pytorch, LLM — *Philadelphia, PA*
- Expanded a novel Bayesian inspired neural network initialization algorithm leveraging co-occurrence and Product Quantization, improving pre-gradient update accuracy from ≈10% (random) to 65–74% on CIFAR-10 and MNIST
- Validated the approach through weight evolution analysis, showing 76% cosine similarity between initialized and fully trained kernels, demonstrating its effectiveness as an informative warm-start for neural networks
- Integrated RLHF with 1.5B parameter Language Model via preference-based fine-tuning with DPO and adaptive training, improving performance by 17% and reducing perplexity from 31.64 to 25.42 on HHH dataset

Precisely — July 2022 – June 2023
Associate Software Engineer - Location Intelligence R&D dept. | Python, SQL — *Bangalore, India*
- Enhanced Geocoded Master Location Data pipeline, expanding coverage to Sweden and France while optimizing data processing for 1M+ geographic locations
- Productionized World Fabric Data automation pipeline by integrating UPU postal data, reducing processing time by 67%, improving cross-country data consistency and reducing manual validation effort
- Automated QA framework, streamlining test case execution, identifying potential discrepancies and rectifying issues within the Alternate Language Data over various countries, resulting in a 28% improvement in accuracy

Precisely — Jan 2022 – June 2022
Intern - Location Intelligence R&D dept. | TensorFlow, SQL, Python — *Bangalore, India*
- Debugged and optimized 50+ SQL queries and validation pipelines across multiple international postal systems (Finland, India, China/HK), improving ETL accuracy by 6% and reducing validation time by 14%
- Implemented address deduplication PoC using Bi-LSTM with GloVe embeddings, achieving 86% accuracy in identifying duplicate addresses across diverse formats
- Contributed to a large-scale data cleaning and transformation pipeline processing 1M+ addresses, reducing duplicates by 40%, and collaborating with data science teams to ensure data consistency and reliability at scale

Projects & Research

Synthetic Text Data Generation Pipeline with LoRA [Link] | *Pytorch, LoRA* — June 2025 – July 2025
- Designed and implemented an end-to-end machine learning data pipeline (ingest, validation, cleaning, monitoring) for synthetic text generation across AG News and IMDB datasets, enabling zero, one, and few-shot text generation
- Orchestrated workflows with Prefect, containerized with Docker + CI/CD, and exposed the pipeline through RESTful APIs (FastAPI) for integration into downstream systems
- Integrated logging using MLflow and monitoring to track generation quality and diversity (ROUGE-L, Self-BLEU, N-gram div.), ensuring continuous evaluation and reproducibility across 10+ tunable hyperparameters

LLaMA Vector Search Engine [Link] | *LangChain, Retrieval-Augmented Generation* — Aug 2024 – Sept 2024
- Architected AI research assistant using LangChain and LLaMA 3.2, enabling semantic search across 30+ ML research papers with Mean Reciprocal Rank of 86%
- Implemented vector-based document retrieval system using FAISS and OllamaEmbeddings, reducing search latency by 56% while processing 5-6 chunks/second
- Optimized search accuracy using Maximal Marginal Relevance (MMR) based retrieval system, improving result relevance by 27%, supporting parallel query processing and documented technical details for reproducibility

Technical Skills

Programming & Dev Tools: Python, C/C++, Java, NumPy, Pandas, SQL, Git, Docker, Spark, AWS, Azure, Jira
ML/AI Tools: Scikit-Learn, PyTorch, TensorFlow, OpenCV, LlamaIndex, HuggingFace, LangChain, ONNX, MLflow
LLMs & Gen AI: Models (LLaMA, Mistral, OpenAI, Deepseek), LoRA, Finetuning, RAG, FAISS, Pinecone
Certifications: Introduction to Machine Learning using TensorFlow - Udacity | Data Science for Engineers - NPTEL
Achievements: Location Intelligence Hackathon Runner up (Precisely), CCI Dean's Fellowship (Drexel)